

Reliability of Microscaling-Format Neural Networks under Bit Faults

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1 Background and theory

Low-precision inference is limited by dynamic range: a 4- or 6-bit number cannot span the spread of values in a weight tensor. Microscaling (MX) formats solve this by sharing one scale across a small block. A block of K elements is stored as one 8-bit exponent X (the E8M0 scale) together with K narrow element codes m_i , and the hardware reconstructs

$$v_i = m_i \cdot 2^{X-127}, \quad i = 1, \dots, K. \quad (1)$$

An element code is a small sign-exponent-mantissa number decoding to $(-1)^s(1+f/2^M)2^{e-\text{bias}}$ for $e > 0$; e4m3 uses 1+4+3 bits, e3m2 1+3+2, e2m1 1+2+1. The two 8-bit formats reserve codes for NaN and infinity, whereas the 6- and 4-bit formats have none, so every one of their bit patterns is a finite number. That distinction turns out to matter more than any other. The scale follows the OCP rule $X = \lfloor \log_2 \max_i |v_i| \rfloor - e_{\max}$, with $e_{\max}$ the exponent of the largest value the element format can hold.

In ordinary floating point a value is a self-contained word, so a bit flip perturbs one number. Equation (1) breaks that. An *element* fault flips a bit of one m_i : a mantissa bit changes the value by at most a relative 2^{j-M} , an exponent bit multiplies it by $2^{\pm 2^j}$, and the sign bit negates it. A *scale* fault flips a bit of X , so $X \rightarrow X \pm 2^j$ and all K elements are multiplied by $2^{\pm 2^j}$ at once; if the flip produces the reserved code 255 the whole block becomes NaN. The number of values a fault alters is its *blast radius*: 1 for an element fault, K for a scale fault.

This predicts something before any measurement. A block occupies $8 + Kw$ bits (w = element width), of which only 8 belong to the scale, about 3% for $K = 32, w = 8$. Yet if every stored bit is equally likely to flip, the expected number of corrupted *values* per block is $8K$ from scale faults against Kw from element faults, a ratio of $8 : w$. An 8-bit format therefore takes about half its corrupted values through a scale occupying 3% of storage, and a 4-bit format two thirds.

2 Fault model and metrics

I use the single-bit-flip model standard in fault-injection work, applied to stored values: one bit of one element code, or of one scale byte, is inverted. Weight faults are persistent and affect every inference; activation faults are transient and affect only the inference in which they occur. Let $D = \{(x_i, y_i)\}_{i=1}^N$ be a fixed evaluation set, $\hat{y}_i$ the prediction of the fault-free but MX-quantised network, and $\hat{y}_i^\varphi$ the prediction under fault φ .

metric	definition	meaning
corruption count	$c(\varphi) = \sum_i \mathbb{K}[\hat{y}_i^\varphi \neq \hat{y}_i]$	inferences the fault changed
SDC indicator	$s(\varphi) = \mathbb{K}[c(\varphi) \geq 1]$	did it corrupt anything at all
SDC rate	$\hat{p} = \Phi ^{-1} \sum_{\varphi \in \Phi} s(\varphi)$	fraction of faults that corrupt
per-inference rate	$r(\varphi) = c(\varphi)/N$	chance a given inference is corrupted
non-finite rate	$ \Phi ^{-1} \sum_{\varphi} \mathbb{K}[\text{NaN/Inf output}]$	faults that destroy the output

The SDC rate counts a fault as a failure if it changes even one prediction, so it is deliberately sensitive and comparable across configurations rather than across papers. Weight and activation faults must not be compared on it, since a weight fault is exposed to all N inferences and an activation fault to one; the comparable quantity is $r(\varphi)$. Rates carry Wilson 95% intervals. Because the element alphabets hold at most 256 codes, the decoded value of every (code, bit) pair is tabulated exactly rather than sampled.

3 Method

The campaign compares a golden run against a faulty one. The network is first put in the form an accelerator executes: batch normalisation folded into the preceding convolution, and RepVGG's branches fused. This matters because an unfolded batch norm re-normalises whatever the corrupted convolution produced and hides part of the damage. Weights are then encoded into real MX bit patterns, the fault is applied *in the code domain*, and the block is decoded only afterwards, which is what lets a scale fault perturb all K values.

*Replace with your full name and roll number.

Algorithm 1 Fault-injection campaign

Require: network M , format F , block size K , evaluation set D , budget n

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1:  $M \leftarrow \text{DEPLOY}(M)$  ▷ fold batch norm, fuse branches
2: for all quantisable layers  $\ell$  do
3:   view  $W_\ell$  as (outputs  $\times$  reduction), split the reduction axis into blocks of  $K$ 
4:    $X_\ell \leftarrow \lfloor \log_2 \max |W_\ell| \rfloor - e_{\max}$  per block;  $C_\ell \leftarrow \text{round}_F(W_\ell 2^{-X_\ell})$ 
5:    $W_\ell \leftarrow \text{DECODE}(X_\ell, C_\ell)$  ▷ the network now holds MX values
6: end for
7:  $\hat{y} \leftarrow \arg \max M(D)$  ▷ golden predictions
8: for  $t = 1$  to  $n$  do
9:   draw  $\varphi = (\ell, \text{site}, \text{index}, b)$  uniformly over all stored bits
10:  if site = element then  $C_\ell[\text{index}] \leftarrow C_\ell[\text{index}] \oplus 2^b$ ; update that one weight
11:  else  $X_\ell[\text{index}] \leftarrow X_\ell[\text{index}] \oplus 2^b$ ; update the  $K$  weights of the block
12:  end if
13:   $\hat{y}^\varphi \leftarrow \arg \max M(D)$ ; record  $c(\varphi)$ ,  $s(\varphi)$ , non-finiteness; restore weights
14: end for
15: return rates over  $\Phi$  with Wilson intervals
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Line 8 defines the sampling. Drawing bits uniformly is the physically faithful choice, since faults land in proportion to storage; replacing that line by an enumeration of every addressable bit makes the campaign *exhaustive*, giving the true failure rate with no sampling error.

Setup. CIFAR-10 with ResNet8 (78k parameters, 87.0% top-1), RepVGG-A0 (7.0M, 91.4%) and a vision transformer (2.7M, DeiT-Tiny width, 81.9%), plus ResNet8 at five widths from 20k to 692k with ten independently trained models each for the capacity study. Formats e4m3, e5m2, e3m2, e2m3, e2m1, block sizes 8–64. The evaluation set is a fixed class-balanced subset of 200 test images; campaigns use 3000 injections unless stated otherwise.

4 Results

Exhaustive ground truth. Flipping all 638,624 bits of ResNet8’s MXFP8 weight storage gives a true SDC rate of 0.4171. The ordinary 3000-injection campaign returned 0.4270 with interval [0.4094, 0.4448]: 2.4% off in relative terms, interval containing the truth. The sampling the rest of the study relies on is therefore verified, not assumed. Exactly, element faults corrupt at 0.4003 and scale faults at 0.9442, with 3.27 against 49.89 of 200 inferences corrupted. Two scale bits are catastrophic without exception, both at rate 1.000 over 2468 flips each: bit 7, which shifts the exponent by 128 and leaves the float32 range, and bit 3, worth a factor 256. Within an element code the sign bit is the most damaging position of all, at 0.593. Vulnerability runs opposite to layer size: the 3584-bit stem is the most fragile layer at 0.853, the 304,128-bit final convolution the most robust at 0.289, so a uniform campaign spends 48% of its budget on the most robust layer.

Accuracy does not predict reliability, for two separate reasons. On the transformer every format except MXFP4 lands within 0.2 points of fp32 accuracy, so e5m2 and e3m2 differ by 0.01 points – and their element SDC rates are 0.0730 ± 0.0089 and 0.0041 ± 0.0047 , a factor of 17.8. Two independent mechanisms produce such gaps.

Special codes. Only formats owning NaN/Inf codes can produce a non-finite value, and nothing downstream absorbs one. On the transformer 94% of e5m2’s element failures were non-finite events, against 27% for e4m3 and necessarily none for e3m2. Training ResNet8 at five widths, ten seeds each, isolates this: the non-finite rate is a property of the format, varying under 20% across a 35-fold change in capacity with no correlation against model size, while the benefit a format draws from capacity falls with its number of special codes.

format	NaN/Inf codes	capacity slope	sensitivity s (median)
e3m2	0	-1.095 ± 0.098	0.0121
e4m3	2	-0.705 ± 0.038	0.1249
e5m2	8	-0.495 ± 0.032	0.0122

Table 1: Capacity slope is $d \log(\text{SDC}) / d \log(\text{params})$ over ten seeds, differences paired within a seed; steeper means capacity helps more. Sensitivity is defined in the next paragraph.

All three pairs separate: e3m2 against e5m2 gives -0.600 ± 0.075 ($t = -8.01$, negative in all ten seeds), and the adjacent pairs -0.390 ± 0.097 and -0.210 ± 0.052 ($t = -4.00$ and -4.01).

Mantissa width, through sensitivity. A second mechanism appears among formats with no special codes at all. Perturbing a single weight by a fixed multiple of the layer RMS – a probe containing no bit patterns –

separates the quantised networks completely: at 1.5 RMS the 2-mantissa networks never flipped a prediction in 1500 trials, while the 3-mantissa ones flipped 2–5% of the time. The fragility therefore belongs to the network, not to the encoding. Its cause is the sensitivity $s_i = |\partial m / \partial w_i| \text{rms} / m$, where m is the margin between the top two logits: a perturbation of $\delta \cdot \text{rms}$ should flip a prediction when $s_i \delta > 1$, and computing s by backpropagation reproduces the measured probe with correlation +0.998. Mean margins, logit spreads and mean gradients agree across formats to three digits; only the *tail* of s differs, by a factor 14 at the 99th percentile. Coarse mantissas suppress that tail, so the finer format is the fragile one – the opposite of what per-fault severity suggests.

Shared scales dominate on every architecture. Scale faults outweigh element faults by 2.4× on ResNet8 (exhaustive), 17.8× on RepVGG-A0 and 12.1× on the transformer for e4m3, and they corrupt 16–50 of 200 inferences each. They also produce non-finite outputs at 7–10% *for every format*, including those with no special element codes, because the E8M0 scale has its own NaN code and an MSB worth 128 exponent steps. Protecting the shared scale pays on convolutional, re-parameterised and attention architectures alike, and in both tensor types: on the per-inference rate, element faults are 5.3 times more damaging in weights than activations, while scale faults are about equally dangerous in both.

Block size, and how much a single model can be trusted. Raising K from 8 to 64 cuts the storage share of scale bits from 11.1% to 1.7% while a scale fault’s SDC rate rises from 0.731 to 0.941, with accuracy nearly flat in K : a clean efficiency-against-reliability trade. Across ten seeds per width, the spread between independently trained models is about ten times the uncertainty contributed by 3000 injections, so a reliability figure quoted from one trained model is far less certain than its injection count suggests.

5 Two problems with the proposed methodology

The severity measure, and a replacement. The plan scores an element fault by $d = |\log_2 |v'| - \log_2 |v||$. Over 39,337 element faults its rank correlation with observed SDC is -0.035 and its AUC is 0.435 – it ranks failures slightly *worse* than chance. The reason is visible in the definition: negating a value leaves $|v|$ unchanged, so a sign flip scores zero severity although the exhaustive data show the sign bit to be the most damaging element bit. Three successively better measures follow: relative error (0.699), absolute change in layer-RMS units (0.845), and $\text{margin_shift} = s_i |v' - v| / \text{rms}$, the fraction of the decision margin a fault consumes, at AUC 0.990. Its threshold needs no tuning: flagging faults with $\text{margin_shift} > 1$ catches 98.5% of all failures while flagging 4.4% of the fault space, which is directly usable for the value-aware sampling the plan wanted. It also carries a caveat the original formulation hides: severity is not a property of the number system, since the same flip in the same format behaves very differently in two networks of equal accuracy.

The standard scale rule manufactures a trend. Because the largest representable element lies below the top of the binade the OCP rule aligns to, the block maximum is clipped, and smaller blocks mean more maxima and more clipping. Injecting an identical set of faults at every block size, element-side SDC varies by a factor 2.21 on ResNet8 and 7.25 on RepVGG-A0 under the OCP rule, but is flat (1.04 and 1.48) under a rule that cannot clip. The apparent best block size for reliability is an artefact of the scale rule, so every result obtained under it needs a non-clipping control.

6 Status

The study covers three architecture families and answers all five objectives. Design guidance follows directly: prefer element formats without NaN/Inf codes, expect coarser mantissas to be more fault-tolerant at equal accuracy, and protect the shared scale before anything else. The gap is scale – everything is CIFAR-10, because ImageNet is not available on the hardware I can reach. One question stays open: why coarse quantisation suppresses the sensitivity tail. The measurements localise it precisely, but the cause is not yet established.